

Q-I: Answer precisely in the space given. No steps or explanations required.
[30 min; $10 \times 3 = 30$ points]

1. A set of $\mathcal{X} = \{x_i\}$ of N samples $x_i \in \mathbb{R}^d$, come from a multivariate Gaussian. Covariance matrix of $\mathcal{X}$ is $\Sigma = ?$ (write below the expression for computing covariance matrix)
2. A set of data samples $\mathcal{X} = \{x_i\}$ in $\mathbb{R}^2$ (i.e., $x = [x^1, x^2]^T$) to be linearly transformed to a new set $x' = \{x'_i\}$ also in $\mathbb{R}^2$, such that the first feature/attribute is the sum of two old features/attributes (i.e., $x^1 + x^2$) and second one is the difference of the two old (i.e., $x^1 - x^2$). If the transformation is $x' = Wx$, write the matrix W .
3. A set of data samples $\mathcal{X} = \{x_i\}$ in $\mathbb{R}^2$ has mean $[1, 1]^T$. We need to make all the samples unit norm (L2) and also mean zero (i.e., centered). Write python code snippet to this transformation (use numpy if required).
4. We use KNN for a binary classification problem. “If we increase K , the decision boundary is becoming more and more smooth” Is this statement “TRUE” or “FALSE”?
5. You know the MSE loss popular in linear regression as

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (y_i - w^T x_i)^2.$$

Write the gradient descent based update equation for obtaining w^{k+1} from w^k .

6. We are interested in finding the optimal solution to an overdetermined system of homogenous equations $Ax = 0$, where A is $m \times n$, $b \in \mathbb{R}^m$, and $m > n$. The solution is defined as: (choose the right option from below)
 - (a) Eigen vector corresponding to the largest eigen value of AA^T .
 - (b) Eigen vector corresponding to the smallest eigen value of AA^T .
 - (c) Eigen vector corresponding to the largest eigen value of $A^T A$.
 - (d) Eigen vector corresponding to the smallest eigen value of $A^T A$.

7. Consider a simple diagonal matrix A with all diagonal elements a , i.e., $A = aI$. What could be the SVD (i.e., UDV^T) of A ? (Guess. Use intuition and not by numerically computing).
8. Consider a simple 1D dataset (vector of values) $\{(+1, +), (+2, +), (+3, -), (+4, -)\}$. What will be the SVM decision rule and which are the support vectors?
9. We have one dimensional samples each from two normal distributions $\mathcal{N}(20, \sigma^2)$ and $\mathcal{N}(30, \sigma^2)$. The optimal Bayesian classifier/threshold will be:
 - (a) closer to 20
 - (b) closer to 30
 - (c) 25, independently of σ^2
 - (d) more than 30
 (choose the right option)
10. (a) Write the objective function that K-Means algorithm minimizes. (b) Is this algorithm guaranteeing global unique solution (i.e, is this a convex optimization)?

Q-II. Answer precisely in the space provided [30 min; $4 \times 7.5 = 30$ points]

1. Bag I contains w_1 white and b_1 black balls. Bag II contains w_2 white and b_2 black balls.
 A ball is drawn at random from one of the bags, and it is found to be white. What is the probability that it was drawn from Bag I?
 Write all the relevant probabilities and finally compute the required one using Bayes theorem. (Write it as an expression in terms of w_1, b_1, w_2 and b_2)
2. We know that a kernel $k(P, Q) = \phi(P)^T \phi(Q)$. Consider a data set of N samples $\{x_i\}$ and a Kernel Matrix K such that (i, j) -th element of this matrix is $k(x_i, x_j)$.
 - (a) Show that K is symmetric.
 - (b) Show that Kernel matrix (i.e., K) is PSD (Positive Semi Definite).
3. Consider a function $f(w) = w^2 + 2w + 1$. We want to minimize $f(w)$ using gradient descent. Initialize $w^0 = 5$. Choose the learning rate $\eta = 0.1$.

- (a) Show two iterations, i.e., compute the value of w after first and second iterations (w^1 and w^2).
 - (b) Suggest a learning rate better than 0.1 so that it converges faster.
 - (c) Suggest a learning rate that will take you to the optimal solution in one step (i.e., w^1).
4. Consider a simple linear regression problem with $x_i \in \mathbb{R}$ and $y_i \in \mathbb{R}$. We would like to learn a model of $y_i = w_1 x_i + w_2$.

Given the data D of $N = 4$ samples:

$$\{(1, 2), (2, 3), (3, 4), (4, 5)\}$$

Without numerically computing, guess/find $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ by observing the data.

Assume a generic linear regression model:

$$y_i = \theta^T x_i + b, \quad x_i \in \mathbb{R}^d$$

Find θ and b using either a closed-form expression or gradient descent.

Write the Python snippet to do this clearly showing matrix and vector manipulations (do not use direct API calls for linear regression, but you may use numpy).

Q-III. Answer in detail in the space provided [30 min; $2 \times 15 = 30$ points]

1. Consider 10 samples $\{x_1, \dots, x_{10}\}$ (given below) to be clustered into three clusters using K-Means algorithm.

Algorithm is initialized with assignments:

$$\text{Cluster-1} = \{x_1, x_4, x_7, x_{10}\}, \quad \text{Cluster-2} = \{x_2, x_5, x_8\}, \quad \text{Cluster-3} = \{x_3, x_6, x_9\}$$

- (a) Write the steps of K-Means with appropriate equations.
- (b) Numerically show K-Means for 3 iterations (show means and clusters in the three iterations).

The data points are:

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 4 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ 6 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ 4 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 6 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -6 \\ -4 \end{bmatrix}, \begin{bmatrix} 0 \\ -4 \\ -6 \end{bmatrix} \right\}$$

2. We know PCA (or Eigenfaces) can be used for dimensionality reduction (say from 10^4 to 10^2) and then use a simple classifier for tasks like face recognition.

We are given a training set of 100 face examples each of two individuals “A” and “B” (i.e., 200 face images), each of size 100×100 (first 100 are from “A” and second 100 are from “B”).

We are given a new test image Q of size 100×100 . Decide whether this face is of “A” or “B”.

Write the precise equations/steps and Python snippets (use numpy if required) for:

- (a) Computing mean and covariance
- (b) Computing and choosing the right eigenvectors
- (c) Reducing dimensionality of training data from 10^4 to a lower appropriate dimension d
- (d) Performing K-NN based classification of the test image Q (assume $K = 5$)